Isar Aerospace: Noticing a Billion-Euro Idea in a Student Rocketry Club

Teaching Case (HBS-style) — written for this course

Prepared for: TUM Entrepreneurship Course: Prototyping Entrepreneurial Ideas for AI

Decision Point: Winter 2017/18 — Three Rocketry-Club Engineers, One Noticed Gap, Three Ways to Spend the Next Decade

How this case ties into Module 02

This is the ideas module — and Isar Aerospace is the ideas lecture compressed into one company a few S-Bahn stops from this classroom. As you read, work three connections: (1) **Idea 1 — noticed, not brainstormed:** nobody in this story runs an ideation workshop. The founders lived in a corner of the future — a rocketry club on the Garching campus — and the gap (European satellites with no European ride to orbit) was visible from there years before it made headlines. Ask which corner of the future you already live in. (2) **Ideas 2/3 — diverge, then gate:** the founders' option set and the DVF run in Exhibit B are your Idea Stress Test at 1:1 scale — and the Feasibility column turns out to be founder-relative. (3) **Idea 4 — founder-problem fit:** why THEM (they had built engines), why THIS (they had lived next to the complaint for years), why NOW (the New Space window); this week's Mom Test interviews are how you earn the desirability evidence these founders absorbed by accident. Then, after class, read the epilogue: Module 00's "succeed or fail quickly" ethos, run at €400 million scale, thirty seconds at a time.

Introduction

The most valuable startup idea in Munich in 2017 was not hiding. It met regularly, in a student workshop on TUM's Garching campus, and it wore safety goggles.

WARR — the Wissenschaftliche Arbeitsgemeinschaft für Raketentechnik und Raumfahrt — has existed at TUM since 1962. It is one of the university's oldest student groups, and it does not build slide decks. It builds hardware: rocket engines, test stands, small satellites,

and, for a few famous years, Hyperloop pods that beat every other university on earth in SpaceX's competitions.

Three of its members were Daniel Metzler, who led WARR's roughly 40-person rocketry team; Josef Fleischmann, who developed rocket systems in the club and was part of the WARR team that won the first-ever SpaceX Hyperloop competition; and Markus Brandl. All three were TUM engineering students in their mid-twenties. None had founded a company. What they had was unusual and specific: they had actually built and test-fired the hard parts of a rocket, as students, on a club budget.

And from inside that club, they could see something the European aerospace establishment was not acting on: satellites were getting small, numerous, and commercial — and Europe had no affordable, flexible way to launch them.

By TUM's own account, the idea took shape in WARR's workshops in 2017. Fleischmann later put it plainly: "In our mid-20s, we had this crazy idea: Why not try building a launch vehicle to transport satellites into space?"

Said like that, it sounds like romance. This case is deliberately not about romance. It is about mechanics: what exactly these three noticed, why they could notice it before better-funded people acted on it, and how anyone should decide whether a noticed gap is a company — let alone one that will need a decade and hundreds of millions of euros before it can succeed even once.

A note on method: the public facts in this case — dates, people, quotes, funding, flight events — are sourced and verifiable. Scenes and internal deliberations are marked as illustrative reconstructions. The scenes are reconstructed; the decision was real.

Background

The club: sixty years of permission to build

WARR was founded in 1962, decades before "startup" was a word anyone at TUM used. Its defining feature is not talent — every technical university has talent — but permission: TUM's own retrospective says the students were given all the freedom they needed to tinker with rocket engines. Members did not study propulsion; they machined it, assembled it, and stood behind concrete when it was ignited. Projects ran from hybrid and liquid engines to small satellites and a space elevator; between 2017 and 2019, WARR's Hyperloop offshoot won SpaceX's pod competition repeatedly, at up to 467 km/h — beating MIT, Delft, and Stanford.

Keep one thing in view: this is a student club. Weekday evenings, volunteer labor, sponsor parts, exam-season absences — exactly the kind of place a "serious" observer would never scan for a billion-euro company.

Living in the future, on a student budget

Module 02's first idea says good ideas are noticed by people who already live where the future is arriving early. In 2016 and 2017, the future of the space industry was arriving early inside WARR — not because the club was secretive, but because its members read launch news the way other students read football scores.

What they saw, all public and verifiable (Exhibit D has the full list): satellites collapsing in size and cost — 104 of them rode a single Indian rocket in February 2017, 88 from one Silicon Valley company; launch being commercialized by startups — SpaceX re-flew a used orbital booster that March, and venture-backed Rocket Lab test-flew its small Electron in May 2017 and reached orbit in January 2018; and constellations turning launch demand from occasional to industrial — OneWeb funded past a billion dollars, SpaceX filing with the US regulator for more than 4,000 satellites.

Anyone could read these facts. The question Module 02 cares about is: who felt them as daily, concrete reality rather than industry news? People who were already building the machinery involved.

The gap: Europe could build satellites, but not launch them on its own terms

Now look at the same years from the perspective of a European team with a finished 150-kilogram satellite.

Europe's launch capability was built for a different era. Ariane 5 was a heavy-lift vehicle, superb at placing large telecom satellites and institutional payloads — and priced accordingly. Vega, flying since 2012, served the roughly 1.5-tonne class with an institution-heavy manifest. Ariane 6, announced in 2014, was targeted for 2020. None of it was designed for the new customer: a commercial team with a small satellite, a specific orbit, and a board asking why deployment was eighteen months late.

In practice, small European satellites flew as rideshare — secondary passengers on someone else's rocket, on someone else's schedule, to someone else's orbit, frequently on non-European vehicles. If the primary payload slipped, you slipped. There was no dedicated European small-launch provider. In 2018 this read mostly as a commercial-flexibility problem; the sovereignty framing that dominates today's headlines came later, and it made the gap larger, not smaller.

Metzler summarized the founders' read years afterward: "Even before it was obvious that relying on a single company or nation was risky, we trusted our own understanding of the market… demand for more launch capacity in Europe and the rest of the world was huge from the start."

The noticing

The following paragraph is an illustrative reconstruction of the pattern — consistent with the founders' later public accounts, but the specific conversations are invented for teaching purposes.

The signal did not arrive as a market report. It arrived as repetition. Student satellite teams and early-stage space companies in the club's orbit kept circling the same question: the satellite is the easy part — how do we actually get it up? At conferences, the corridor conversation repeated: manifests full, rideshare slots late, orbits wrong, prices opaque. Heard once, that is an anecdote. Heard for the tenth time by someone who has personally built an engine, it becomes something else: a gap with a shape.

This is worth stating precisely, because it is the whole module in one sentence: **the founders did not generate an idea; they took seriously a complaint they had been standing next to for years.**

Munich, 2018

The trio incorporated Isar Aerospace in Munich in 2018 — named after the river — at the center of an ecosystem built for exactly this handoff: TUM as the technical base, WARR as the proving ground, and UnternehmerTUM's orbit providing early support, with its affiliated fund UVC Partners joining the seed round alongside Munich's Vito Ventures and Bulent Altan, a former SpaceX vice president who would later become Isar's chairman. In December 2019, Earlybird and Airbus Ventures led a $17 million Series A.

But that is the ending of a different chapter. The decision this case examines came before any of it.

The Situation

Three engineers, one table

(Scene reconstructed; profiles factual.)

Winter 2017/18. Three WARR members in their mid-twenties, degrees finishing or freshly finished, sitting with the question the club's workshops had quietly assembled for them.

What they brought was easy to underrate and hard to replicate: Metzler had run a 40-person volunteer engineering organization — recruiting, planning, shipping hardware without paying anyone; Fleischmann had built rocket systems and won the most visible student engineering competition in the world; all three had hands-on rocketry experience that virtually no European founder outside the institutional aerospace world possessed. They knew from direct experience what a small team could build for how much money — a number orders of magnitude below what the incumbent system assumed.

What they did not have: capital, customers, launch experience, or any evidence that private investors would fund a European rocket company at all.

The three paths

(The option framing is a teaching reconstruction of the choice the founders faced; the facts inside each option are real.)

Option A: Join the establishment

Take the Ariane path — a career at one of Europe's aerospace institutions or primes: the default trajectory for top WARR graduates, and the risk-free way to work on rockets.

For: Real rockets, real salary, real pension. Europe's institutional programs were where the continent's launch expertise actually lived. If the small-launch market materialized, the incumbents might build for it — from inside, you could be part of that.

Against: The gap existed because of how the institutional system worked — optimized for heavy payloads and long cycles, not for a startup customer with a 150 kg satellite and a deadline. Joining it meant betting the system would fix the blind spot it had produced. And the window was open now: SpaceX and Rocket Lab had just made private launch investable.

Option B: Found the launch company

Build a privately financed European launch vehicle for small and medium satellites — the company that became Isar Aerospace.

For: The gap was real and repeatedly confirmed by the people living it. The founders held the single rarest asset for this idea: demonstrated ability to build propulsion hardware cheaply. The demand curve pointed one direction. No dedicated European competitor existed yet. If it worked, the prize was not a business — it was the European launch franchise for a satellite generation.

Against: Everything else. Development costs for comparable vehicles ran into nine figures (Exhibit C). First revenue would be years away; the feedback loops Module 07 will teach you to compress are, in launch, measured in years and ended by explosions. No privately developed European rocket had ever reached orbit. And three students would be asking European investors — famously hardware-shy in 2018 — to fund a decade-long bet against physics, ArianeGroup, and SpaceX simultaneously.

Option C: Take the sensible startup idea instead

Found something fundable and fast: B2B software, industrial AI, a marketplace — 2018 Munich was full of capital-light paths where feedback arrives in weeks, not years.

For: Faster validation, smaller cheques, more forgiving failure modes, an ecosystem tuned for it. By every startup-handbook heuristic, the rational first company.

Against: Founder-problem fit near zero. Their decisive advantage — built engines, led hardware teams — would be worth nothing; in software they would be ordinary founders among thousands. And noticed gaps do not wait: they get taken, usually by whoever else was standing in the future.

The gate

Run Module 02's DVF filter on Option B as the founders might have in early 2018 (Exhibit B does this in full). **Desirability:** strong signal, weakly proven — satellite operators visibly suffered; nobody had yet signed a euro of intent. **Viability:** the honest hole — would customers pay enough, at a cadence high enough, to sustain a privately financed European launcher? Unproven anywhere in Europe. **Feasibility:** their edge, and the inversion that makes this case worth teaching — for almost any founding team on earth, "build an orbital rocket" fails feasibility instantly; for this team it was the strongest column, because they had already built what everyone else considered the impossible part.

Which exposes the question your Idea Stress Test glosses over if you let it: DVF scores are not properties of an idea. They are properties of an idea in specific hands, at a specific time.

The ambition question

One more thing was clear even in 2018: this idea had no small version. Launch is a fixed-cost, minimum-scale business — you cannot launch a third of a rocket to a third of orbit for a third of the money. Exhibit C walks the "ambition ladder" from consulting to components to brokerage to full launch provider, and shows why the middle rungs are strategically unstable. If the three of them acted on this gap at all, they were choosing the top rung: venture-scale, capital-intensive, winner-take-most. Ambition here was not a temperament; it was a structural requirement of the idea.

The Decision Point

It is early 2018, in Munich. Three engineers, one noticed gap, three paths.

Core question: If you are Metzler, Fleischmann, and Brandl, which path do you take — and on what evidence, gathered how cheaply, would you change your mind?

Underneath it, the module's test, applied honestly:

- **Why you?** Answer it for all three options, not just the flattering one. (Option A's honest answer: "because we are exactly who they hire." That is also fit.)
- **Why this?** Is a ten-times-heard complaint from your own environment better or worse evidence than a market report — and what would make it paying evidence?
- **Why now?** What specifically about 2018 — not 2013, not 2023 — made this the window?

Questions for Deliberation

1. **Information versus conviction:** Every fact about the launch gap was public in 2017 — reports, panels, launch manifests. What did three WARR members have that an analyst holding the same facts did not? Separate information advantage, conviction advantage, and cost-of-testing advantage — which of the three actually built Isar?
2. **Founder-relative DVF:** Run Exhibit B's gate twice — once as the WARR trio, once as two business students with identical market data. Which cells change, and by how much? What does that do to the idea of a "good idea" as an absolute category — and to how you read your own team's DVF scores this afternoon?
3. **The safer-idea trap:** Option C looks lower-risk: less capital, faster feedback. Argue the opposite — that for these three founders specifically, the software startup was the riskier path. What is the expected cost of holding a rare edge and not spending it?
4. **The viability hole:** In 2018, no privately developed European rocket had reached orbit, and Europe's flagship launcher operated inside an institutional support system rather than as a standalone commercial business. What was the cheapest credible test of "customers will pay, at prices that work"? Letters of intent? A paid design study? And what result would you have accepted, in writing, as a kill signal?
5. **The ambition ladder:** Could they have entered one rung down — propulsion supplier to the primes, or launch brokerage — earned revenue early, and climbed? Use Exhibit C: which rungs are stable businesses, which are waiting rooms, and why does the launch market structurally punish the middle?

6. **Why now:** Same team, same idea, founded in 2013 or in 2023 instead of 2018. Walk each timeline: capital climate, demand curve, competitor field, incumbent programs. Which single factor does the 2018 window actually hinge on?

Exhibits

Exhibit A: Where the idea came from

Timeline of verified events; one row of interior color is explicitly marked as illustrative reconstruction.

When	What happened	Status
1962	WARR founded at TUM — one of the university's oldest student groups; builds rocketry hardware, not concepts	Verified
2015–2017	The future founders are active in WARR: Metzler leads the ~40-person rocketry team; Fleischmann develops rocket systems and is part of the team winning SpaceX's first Hyperloop competition (the club wins repeatedly through 2019, pods up to 467 km/h)	Verified
2016–2017	The New Space signal accumulates in public view: 104 satellites on one rocket, first booster re-flight, a startup flying, constellations funded in the billions (Exhibit D)	Verified
~2016–2017	Recurring pattern around the club: satellite builders keep asking how they would actually get to orbit; the same rideshare complaints (schedule, orbit, dependence on non-European vehicles) repeat in project work and conference corridors	Illustrative reconstruction — pattern consistent with the founders' later accounts; specific conversations invented
2017	The idea takes shape in WARR's workshops (TUM's own retrospective dates it here)	Verified
2018	Isar Aerospace founded in Munich by Metzler, Fleischmann, and Brandl; seed backing includes UVC Partners (UnternehmerTUM-affiliated), Vito Ventures, and former SpaceX VP Bulent Altan	Verified
Dec 2019	Series A: $17 million, led by Earlybird and Airbus Ventures	Verified

The teaching shape of this table: **years of standing in the future → a repeating complaint → a noticed gap → a feasibility edge earned by having already built → founding.** Nowhere in it is a brainstorming session.

Exhibit B: The DVF gate, as the founders might have run it in early 2018

Illustrative reconstruction for classroom use — the founders never published such an assessment; scores are teaching estimates. The evidence entries are factual.

	What they could point to (early 2018)	What was genuinely unknown	Score (1–5)
Desirability	Smallsat boom underway (104 satellites on one flight, 2017); constellations funded (OneWeb >$1B); European operators dependent on foreign rideshare with schedule and orbit compromises; recurring complaints in their own environment	Would operators switch and pay, or just complain? Zero signed intent existed	4
Viability	Rocket Lab's pricing signaled a market for dedicated small launch; demand forecast to grow for a decade	Could a European entrant reach a cadence and price that sustains a standalone business? No private European precedent	2–3
Feasibility	They had personally built and test-fired rocketry hardware on a student budget; SpaceX and Rocket Lab had proven private teams can do launch; Munich supplied talent via the TUM/ WARR pipeline	Scaling from club hardware to an orbital vehicle: roughly 100x in complexity and capital	3 — **but 1 for almost any other founding team**

Two readings for class: (1) the profile is the opposite of a typical student idea, which scores high on feasibility ("we can build an app") and hollow on desirability; (2) the bracketed note in the Feasibility cell is the case's core claim — DVF is founder-relative. The same idea, gated by a different team, fails.

Exhibit C: The ambition ladder — why this idea had no small version

Analytical teaching device — the ladder framing is ours, not the founders'. Benchmark figures are publicly reported.

Rung	The business	Capital to enter	Ceiling and who captures the value	Stable?
1. Engineering consultancy	Sell rocketry expertise to primes and agencies	~€0	Billable hours; clients capture the upside	Stable but small
2. Component supplier	Sell engines/ subsystems to incumbent launcher programs	€1–10M	Capped by primes' procurement; your innovation prices into their product	Conditionally stable, dependent
3. Launch broker / rideshare aggregator	Resell and package others' launch capacity	€1–5M	Margin on someone else's scarce asset; squeezed when supply integrates forward	Unstable
4. Launch provider	Build and operate the vehicle	€100M–500M+ before first revenue	Own the scarce asset; constellation-era demand; set price and schedule	Winner-take-most, or nothing

Public reference points for rung 4, visible in 2018: Falcon 1 plus Falcon 9 development was publicly estimated near **$400 million**; Rocket Lab had raised roughly **$150 million** before Electron reached orbit; Electron's development was reported on the order of **$100 million**.

Why the market punishes the middle: launch is a fixed-cost cadence game. Each flight amortizes the factory, the pad, and the engineering organization; higher cadence lowers price per kilogram, which wins the manifest, which funds cadence. That loop concentrates each segment around one or two providers and starves half-committed entrants. So acting on this gap meant rung 1–2 (keep your job, effectively) or rung 4 (venture-scale, a decade, hundreds of millions). The idea dictated the ambition level — which is exactly the question to ask of your idea in today's workshop: what is the minimum ambition at which it is a business at all?

Exhibit D: What the future looked like from inside the club, 2016–2018

All entries verified public events.

Date	Signal	Why it mattered, seen from Garching
Nov 2016	SpaceX files with the US regulator for a constellation of 4,400+ satellites	Launch demand becomes industrial, not occasional
Dec 2016	OneWeb raises over $1B (SoftBank-led)	Constellation demand is funded, not hypothetical
Feb 2017	Indian PSLV orbits 104 satellites in one launch, 88 from Planet	Satellites are a product line; rideshare is the bottleneck
Mar 2017	SpaceX re-flies an orbital-class booster (SES-10)	The launch cost curve bends; private launch economics validated
May 2017	Rocket Lab's Electron makes its first test flight	A venture-backed startup — not a state — flies an orbital-class rocket
Jan 2018	Electron reaches orbit	The category "private small-launch company" now exists
Same period, Europe	Ariane 5 serves heavy/institutional payloads; Vega (~1.5 t class) flies an institution-heavy manifest; Ariane 6 targets 2020; no dedicated European small-sat launcher	European smallsats ride foreign vehicles, on others' schedules, to others' orbits — the gap

The discussion prompt hiding in this table: every row was public. The founders' edge was not access — it was position. They stood where these rows were felt as daily reality, and they had the hands to act on them.

Questions for Class Discussion

1. **The Tuesday-evening inventory:** Isar's founding gap was visible from a student club workshop. List the "clubs" you currently stand inside — labs, jobs, Werkstudent roles, hobbies, communities. What complaint have you personally heard at least three times there? Write down three. (This is Idea 1 made operational.)
2. **If ideas are noticed, why brainstorm?** Reconcile "noticed, not brainstormed" with this afternoon's Crazy 8s. What is divergence actually for, if not invention? (Hint: what does volume do to material you have already absorbed?)

3. **Would you have funded them?** You are a Munich seed investor in mid-2018: three mid-twenties engineers, no revenue for years, nine-figure capital needs, ArianeGroup on one side and SpaceX on the other. Make the honest case for the cheque and for the pass. Where does founder-problem fit legitimately beat market logic — and where is it romance with a term sheet?
4. **DVF in deep tech:** Isar's desirability signal was cheap to collect; the feasibility proof would take the better part of a decade and hundreds of millions. When is the DVF gate an honest filter, and when is it false comfort? What is the deep-tech equivalent of Module 04's landing-page test?
5. **Kill criteria for a rocket:** It is 2018 and you are the fourth co-founder. Complete the sentence in writing: "We stop if ___ by ___." Now stress it: the industry's own norm says first launches usually fail and it "takes a few attempts to reach orbit." What does that norm do to your sentence — and how do you write kill criteria that survive in a business whose honest feedback loop is years long?
6. **What is lying around at TUM in 2026?** This company began twenty minutes from this room, inside a club any of you could join this week. Name one gap visible right now from a TUM lab, club, or internship — something practitioners around you complain about repeatedly — that the market has not priced in. Why has nobody taken it seriously yet? Apply the test: why you, why this, why now.

Epilogue (read after class)

Here is what actually happened.

They chose Option B. The money followed the milestones: an undisclosed seed (UVC Partners, Vito Ventures, Bulent Altan); a **$17 million Series A** led by Earlybird and Airbus Ventures (December 2019); a **$91 million Series B** led by Lakestar (December 2020) — then the largest European spacetech round — extended by **$75 million** in mid-2021, when Porsche SE also took a stake; a **$165 million Series C** (March 2023), extended in June 2024 by **€65 million** with the NATO Innovation Fund, taking total funding past **€400 million** and making Isar the most heavily capitalized independent New Space company in Europe.

Along the way, the company secured what a launch business actually runs on: exclusive access for up to twenty years to a launch pad at Andøya Spaceport in Norway (April 2021), and victory the same year in the German government's microlauncher competition, worth €11 million via ESA's Boost! programme. Isar built its vehicle — Spectrum, a two-stage, 28-meter rocket powered by ten of its own Aquila engines burning propane and liquid oxygen,

designed to carry up to 1,000 kg to low Earth orbit — developing, manufacturing, and testing almost everything in-house.

Then came the part every module of this course has been preparing you to read correctly.

On **March 30, 2025, at 12:30 CEST**, Spectrum lifted off from Andøya — the first orbital launch attempt by a commercial company from continental Europe (outside Russia). The company had spent the preceding weeks deliberately setting the bar in public: this was a test flight, it carried no customer payload, and the goal was data, not orbit. The flight lasted about thirty seconds. The rocket lost attitude control, was terminated by its flight termination system, and fell into the sea in a controlled manner. The pad survived intact. The investigation later traced the failure to an unintended opening of a vent valve and loss of attitude control at the start of the roll maneuver.

Metzler's statement that afternoon: "Our first test flight met all our expectations, achieving a great success. We had a clean liftoff, 30 seconds of flight and even got to validate our Flight Termination System... Now it's time to analyze all data, learn, iterate and be back on the launch pad as soon as possible." Chairman Bulent Altan — twelve years at SpaceX — added that it "normally can take a few attempts to reach orbit," and that Isar had built "not only a rocket, but an entire rocket production system." Vehicles two and three were already in production on the day vehicle one flew.

Read that as Module 00 taught you. This was not spin, because the success criteria were set before the test, in public, and the test was designed to survive its own failure — no customer payload, pad protected, termination system armed. Succeed or fail quickly at €400 million scale is the same discipline as your kill criteria; the loop is just years long and the tuition is measured in launch vehicles.

The market kept underwriting the bet: **€150 million** from Eldridge Industries (June 2025, at a valuation reported well above €1 billion), selection in ESA's European Launcher Challenge, and a **€270 million Series D** (June 2026) — roughly **€800 million** in total. Over 400 employees across five locations; a factory in Parsdorf near Munich designed for up to forty rockets a year; a manifest through 2028 (ESA, the Norwegian Space Agency's Arctic Ocean Surveillance program, an Astroscale debris-removal mission, and others); a second launch site planned in Canada — and demand that swung, within a year, from almost entirely civil to 60% defense. The gap three students noticed as a commercial inconvenience in 2017 is now called European sovereignty, and it is company strategy.

And the honest ending, because this course does not do romance: as of this writing (July 2026), Spectrum has **not yet reached orbit**. The second flight — "Onward and Upward," the qualification mission carrying Isar's first customer payloads — has been scrubbed repeatedly through the first half of 2026 (a pressurization valve in January, weather and

range issues in March, a suspected pressure-vessel leak in April, off-nominal fluid-system behavior in June) and awaits its next window. Hardware is hard; the decision made in that Munich winter of 2017/18 is still an open bet, running on its ninth year.

One last mechanic, so the lesson lands where it should. The founders were not the only ones who noticed — Rocket Factory Augsburg and HyImpulse, two other German launch startups, were founded around the same time. Noticing was necessary and not sufficient. The differentiating moves were the ones this module hands you: they took a recurring complaint seriously instead of filing it as background noise, they gated it honestly against who they were, and they tested feasibility the way the club had taught them — by building.

The best idea in Munich was lying around at a Tuesday-evening student club. It usually is.

Teaching case written for this course from public sources (Isar Aerospace press releases 2019–2026, TUM press releases, ESA publications, SpaceNews, TechCrunch, European Spaceflight, NASASpaceflight, dpa/heise coverage, and a 2025 Vsquared Ventures interview with Daniel Metzler). Quotes are verbatim from these sources. Exhibits and passages marked as illustrative are reconstructions for classroom use.